P09 Hq Solution

Origin of Matter-Antimatter Asymmetry (HQ v54)

Domain: Particle Physics/Cosmology
Model: PAX L5 Narrow L2 General 27B
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Run: PAX_27B HQ v54 Tokens: 7399 Time: 1292.1s Chain: 7c2a10f275308257b95335f6839c1565159650e399175675822ed873083c14f7 Genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560

Subject: Proposed Resolution to the Baryon Asymmetry Problem via Resonant Leptogenesis with CP-Violating Neutrino Oscillations in the Early Universe

1. Core Hypothesis
The observed baryon asymmetry of the universe (BAU), quantified by $\eta_B \approx 6 \times 10^{-10}$, originates from **Resonant Leptogenesis** driven by the decay of heavy right-handed Majorana neutrinos (N_i) in the early universe. Unlike standard thermal leptogenesis which requires extremely high mass scales ($M_N > 10^9$ GeV) and often conflicts with low-energy neutrino data, this hypothesis posits that the necessary CP violation arises from **quasi-degenerate masses** of the lightest two right-handed neutrinos ($M_1 \approx M_2$). This degeneracy enhances the CP-asymmetry parameter ϵ_{CP} through resonant self-energy corrections, allowing successful baryogenesis at accessible energy scales ($M_N \sim 1\text{--}10$ TeV), potentially within reach of future colliders or precision intensity frontier experiments. The generated lepton asymmetry is subsequently converted into baryon asymmetry via non-perturbative electroweak sphaleron processes before they freeze out.

2. Mathematical Framework

The framework relies on the extension of the Standard Model (SM) by three right-handed singlet fermions N_R, leading to the Type-I Seesaw mechanism.

A. Lagrangian Extension:
$$ \mathcal{L}_{\text{BSM}} = \mathcal{L}_{\text{SM}} - \frac{1}{2} \overline{N_R^c} M_R N_R - \left(\overline{L_L} Y_\nu \tilde{\Phi} N_R + \text{h.c.} \right) $$
Where:
* L_L: SM lepton doublet.
* Φ: Higgs doublet, $\tilde{\Phi} = i\sigma_2 \Phi^*$.
* Y_ν: Yukawa coupling matrix.
* M_R: Majorana mass matrix for N_R.

After Electroweak Symmetry Breaking (EWSB), the light neutrino mass matrix is given by the seesaw formula:
$$ m_\nu \approx -v^2 Y_\nu M_R^{-1} Y_\nu^T $$

B. Resonant Enhancement of CP Asymmetry:
In the limit where $M_1 - M_2 \lesssim \Gamma_{1,2}/2$ (where Γ is the decay width), the CP asymmetry parameter ϵ_1 for the decay $N_1 \to \ell \phi$ is enhanced beyond the standard perturbative bound. The total asymmetry includes vertex and self-energy contributions:

$$ \epsilon_1 = \frac{1}{8\pi (Y_\nu^\dagger Y_\nu)_{11}} \sum_{j \neq 1} \text{Im}\left (Y_\nu^\dagger Y_\nu)_{1j}^2 \right f(M_j^2/M_1^2) $$

For quasi-degenerate states ($M_1 \approx M_2$), the loop function $f(x)$ becomes dominated by the Breit-Wigner resonance:
$$ \epsilon_1 \approx \frac{1}{8\pi} \frac{(Y_\nu^\dagger Y_\nu)_{12} \text{Im}(Y_\nu^\dagger Y_\nu)_{12}}{(Y_\nu^\dagger Y_\nu)_{12}^2} \frac{M_1 \Gamma_2}{(M_1-M_2)^2 + (\Gamma_2/2)^2} $$
This allows ϵ_1 to approach $\mathcal{O}(1)$ even if Yukawa couplings are small, satisfying the Sakharov conditions efficiently.

C. Boltzmann Evolution:
The evolution of the number densities n_{N_i} and the lepton asymmetry $n_{L-\alpha}$ is governed by coupled Boltzmann equations including washout effects:
$$ \frac{dn_{N_i}}{dz} = -D_i(z) \left(\frac{n_{N_i}}{n_{N_i}^{eq}} - 1 \right) $$
$$ \frac{dn_{L-\alpha}}{dz} = -\sum_i \epsilon_{i,\alpha} D_i(z) \left(\frac{n_{N_i}}{n_{N_i}^{eq}} - 1 \right) - W_\alpha(z) n_{L-\alpha} $$
Where $z = T/M_1$, D_i represents the decay rate, and W_α represents the inverse decay and scattering washout terms.

3. Proof Sketch / Key Steps

1. **Sakharov Condition Verification**:
* *Baryon Number Violation*: Provided by non-perturbative sphalerons in the SM plasma above $T_c \approx 130$ GeV.
* *C and CP Violation*: Generated by complex phases in the Yukawa matrix Y_ν during N_i decays, amplified by the resonant denominator $(M_1-M_2)^2 + \Gamma^2/4$.
* *Departure from Thermal Equilibrium*: Achieved when the decay rate $\Gamma_{N_1} < H(T=M_1)$ (Hubble expansion rate), ensuring the process occurs out-of-equilibrium.

2. **Derivation of the Resonance Condition**:
* Demonstrate that for hierarchical masses, $\epsilon_1 \propto M_1$. To match η_B, one needs $M_1 > 10^9$ GeV.
* Show mathematically that introducing a splitting $\Delta M = M_1 - M_2$ such that $\Delta M \sim \Gamma$ creates a pole in the propagator correction, boosting ϵ_1 by a factor of $\sim M/\Gamma$.
* Calculate the required fine-tuning: $\delta M / M \sim 10^{-8}$ to 10^{-12} depending on the specific Yukawa texture, arguing this is natural in models with approximate flavor symmetries (e.g., $SU(2)_R$).

3. **Flavor Effects Integration**:
* Incorporate "flavored" Boltzmann equations. At temperatures $T < 10^9$ GeV, charged lepton Yukawa interactions maintain coherence between flavors.
* Prove that flavor correlations can suppress washout rates for specific lepton flavors while preserving asymmetry in others, significantly relaxing the constraints on M_N.

4. **Conversion to Baryon Asymmetry**:
* Apply the sphaleron conversion factor: $n_B = c_{sph} n_L$. In the SM, $c_{sph} \approx -28/79$.
* Integrate the final lepton asymmetry Y_L over the temperature history to yield the final baryon-to-photon ratio $\eta_B = n_B/n_\gamma$.

4. Testable Predictions

If this hypothesis holds, it yields distinct experimental signatures distinguishable from high-scale leptogenesis:

1. **Low-Scale Heavy Neutral Leptons (HNLs)**:
* Existence of sterile neutrinos with masses in the MeV–TeV range.
* **Search Strategy**: Displaced vertices at LHCb, ATLAS, CMS, or dedicated experiments like SHiP, FASER, and DUNE near detectors.

2. **Lepton Flavor Universality Violation (LFUV)**:
* Specific patterns in rare meson decays (e.g., $K \to \pi \nu \bar{\nu}$, $B \to K \mu e$) induced by mixing between active and sterile neutrinos.

3. **Neutrinoless Double Beta Decay ($0\nu\beta\beta$)**:
* While standard seesaw predicts very long half-lives due to tiny m_ν, the low-scale resonant scenario may allow observable rates if the effective Majorana mass m_{ee} receives contributions from the heavy sector interference or if the light neutrino spectrum is inverted/quasi-degenerate.

4. **Cosmic Microwave Background (CMB) Constraints**:
* Precise measurements of the effective number of relativistic species (N_{eff}) and primordial helium abundance (Y_p) could constrain the lifetime and decay modes of these HNLs in the early universe.

5. **Gravitational Waves**:

* If the phase transition associated with the generation of asymmetry involves a strong first-order phase transition (linked to the symmetry breaking scale), it would generate a stochastic gravitational wave background detectable by LISA or DECIGO.

5. Remaining Open Sub-Problems

Despite the theoretical elegance, several critical challenges remain:

1. **Fine-Tuning vs. Naturalness**:
* Achieving the precise mass degeneracy ($\Delta M \sim \Gamma$) required for resonance enhancement without an underlying protective symmetry remains difficult. Is there a dynamical mechanism (e.g., radiative stability) that enforces this degeneracy?

2. **Thermal Corrections and Plasma Effects**:
* Accurate calculation of the thermal widths $\Gamma(T)$ and thermal masses in the dense early universe plasma is computationally intensive. Current approximations may underestimate washout effects or shift the resonance condition.

3. **Connection to Low-Energy Neutrino Data**:
* Mapping the high-energy parameters (Y_ν, M_R) to the measured low-energy observables (m_ν, PMNS mixing angles, Dirac CP phase δ_{CP}) is highly non-trivial. There is a vast parameter space; determining if the specific textures required for resonant leptogenesis are compatible with current oscillation data (especially the normal vs. inverted hierarchy preference) is ongoing.

4. **Non-Perturbative Sphaleron Dynamics**:
* The exact efficiency of the sphaleron conversion depends on the detailed dynamics of the electroweak phase transition. If the transition is a smooth crossover (as suggested by lattice QCD for the SM), the window for converting lepton asymmetry to baryon asymmetry narrows, requiring extensions to the scalar sector (e.g., Two-Higgs-Doublet Models) to ensure a strong first-order phase transition.

5. **Initial Conditions**:
* The solution assumes a thermal initial state. However, inflationary reheating scenarios might produce non-thermal distributions of N_i, altering the Boltzmann evolution entirely. A robust theory must be independent of specific reheating histories.